Freight Custody Verification Framework

An open framework for verifying carrier identity and load integrity at the physical transfer of truckload freight

Version 0.1, draft for public comment

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Why this document exists

Every truckload shipment in the United States changes hands at least twice, at a pickup and at a delivery, and often more than that on multi-stop and relay routes. At almost every one of those handoffs, the person releasing the freight checks a piece of paper against another piece of paper and lets the truck go. Nothing confirms that the tractor at the dock belongs to the carrier the load was given to, and nothing confirms that the seal on the doors is really there, or is the right one, or is still there at the next stop.

That is the gap that deception-based cargo theft exploits. The criminal wins the load with a clean digital identity, often one that passes every check a broker or a vetting service runs, and then relies on the gate not noticing that the truck in front of it does not match. In 2025 the industry recorded $725 million in direct cargo theft losses, up 60% in a year, and the deception-based share has been the fastest-growing part of the problem since 2021. The Federal Bureau of Investigation told the industry in April 2026 to “independently verify shipment requests and pickups using secondary methods prior to releasing any loads.” The Federal Motor Carrier Safety Administration’s new registration system verifies who a company official is, and its own description is candid that it does not verify drivers or physical pickups. The Department of Transportation asked, in a 2025 request for information, which technologies work, what barriers prevent detection, and what data the industry needs to collect.

What none of those documents provides is a shared statement of what a gate should actually do. The industry’s existing standards cover facilities and vehicles. The broker community’s guidance covers vetting at booking and explicitly declines to be a standard. There is no common definition of a verified custody transfer, no common record of what was checked, and no common way to grade the result. Each large operator that has solved part of the problem has done so privately, with equipment and software tuned to its own yards, and the lessons have stayed inside the building.

I wrote this document because I spent two years building and deploying the pieces of that verification inside one national trucking network: a camera-based system that reads the carrier’s USDOT number off the cab at check-in, a detection model that tells whether a seal is physically on the trailer, and the decision rules that let investigators trust both. What I learned is that most of the value is in the process, not the technology. A clerk with a phone, a checklist, and the public FMCSA registry can close most of the gap today. A camera and a model close the rest without adding a minute to the gate. This document writes down the process so that any facility, any carrier, and any broker can follow it, and so that the tools that automate it have a common target to implement.

It is deliberately short, and it is deliberately a draft. The requirements come from practice, not theory, and practice at other facilities will differ from mine. I would rather publish a version that others can correct than wait for one I cannot be corrected on.

1. Scope

This document, referred to below as the framework or FCVF, specifies what a party that releases, receives, or carries truckload freight must do to verify, at each physical custody transfer, that the vehicle taking custody belongs to the carrier of record for the load and that the load’s seal is present and intact, and what record of that verification must be kept. It applies to pickups, to intermediate stops on multi-stop and relay routes, and to deliveries, of full and partial truckload freight moving by highway within the United States, whether the carrier was engaged directly by the shipper or through a broker or other intermediary.

It does not cover ocean containers at marine terminals, which are governed by customs and terminal operator requirements; parcel and small-package networks, where custody transfers are not performed vehicle by vehicle; or the vetting of a carrier at the time a load is booked, which is addressed by existing industry guidance and commercial services and which this document assumes has taken place. It does not specify physical security of facilities or vehicles, for which the TAPA Facility Security Requirements and Trucking Security Requirements exist, and it is written to be used alongside them rather than in place of them.

The document is written for four audiences. Facility operators, including warehouses, distribution centers, manufacturers' shipping docks, and cross-docks, who release and receive loads. Motor carriers, who present for loads and who are themselves the victims of a large part of freight fraud. Brokers and other intermediaries, who engage carriers and are often the first to discover that a pickup was fraudulent. And the vendors, insurers, associations, and public agencies who build tools for, underwrite, certify, or regulate the handoff. Three conformance levels (Section 8) allow a facility with no software and a facility with automated gates to follow the same framework.

2. How to read the requirements

Requirements are numbered and use the key words defined in RFC 2119 and RFC 8174. "Shall" means the requirement is mandatory for conformance at the stated level. "Should" means the requirement is recommended, and a party that does not follow it ought to understand and be able to explain the reason. "May" means the practice is permitted and optional. Each requirement is tagged with the lowest conformance level at which it is mandatory: Basic, Standard, or Full. A requirement tagged Standard is mandatory at Standard and Full. Text marked "Note" is explanatory and is not part of the requirement.

Requirement identifiers are stable across versions. A requirement that is withdrawn keeps its number and is marked withdrawn; new requirements take new numbers. Identifiers are prefixed by section: G for general, P for pickup, S for intermediate stop, D for delivery, E for the evidence record, V for verdicts, R for risk rules, and C for conformance.

3. Terms

Term	Meaning in this document
Custody transfer	The moment at which physical control of a load passes from one party to another: from a releasing facility to a carrier at pickup, from one driver or carrier to another at a relay or intermediate stop, and from a carrier to a receiving facility at delivery.
Carrier of record	The motor carrier that was engaged to transport the load, identified by its USDOT number and legal name as shown on the rate confirmation, bill of lading, or tender. Where a broker engaged the carrier, the carrier of record is the carrier the broker engaged, not the broker.
Presenting vehicle	The tractor, or straight truck, that arrives to take or deliver custody of a load.
Carrier identity marking	The USDOT number and, where present, the carrier name and MC number displayed on the presenting vehicle as required by 49 CFR Part 390.21.
Registry	The Federal Motor Carrier Safety Administration's public record of registered carriers, as made available through its public query services, including the carrier's legal and operating names, operating status, out-of-service status, and authority.
Seal	A device applied to the closure of a trailer or container that must be broken or removed to open the doors, and that carries a unique identifier. Seals are classified by ISO 17712 as indicative (I), security (S), or high security (H). This document applies to all three and to

Term	Meaning in this document
	electronic and RFID seals.
Seal event	Any application, verification, removal, or replacement of a seal, together with the record of it.
Evidence record	The set of data and images retained for a custody transfer, as specified in Section 6.
Verdict	The graded outcome of a verification: release, verify, or hold, as specified in Section 7.
Independent signal	A verification result that would not be affected by the failure or compromise of another signal used in the same decision. A registry lookup and a photograph of the cab are independent; two readings of the same document are not.
Releasing party	The person or system that decides whether a load leaves a facility or passes a stop.
Verifier	Any person, device, or software that performs a check required by this document.

4. Principles

Five principles run through the requirements. They are stated here so that a reader who meets a situation the requirements do not cover can decide what the framework would want.

Verify the thing, not the paper. Documents describe a load; they do not prove who is taking it. Every verification in this framework is anchored to something physical: the marking on the vehicle, the seal on the doors, the registry entry that the marking must resolve to. Where a document and a physical observation disagree, the observation wins and the load is held.

Use what is already at the gate. The framework asks for no hardware. A phone camera, the public registry, and a written procedure satisfy the Basic level. Fixed cameras most facilities already have, and software that reads them, satisfy the higher levels. A requirement that could only be met by buying equipment would exclude the small facilities and carriers at which most freight changes hands.

Absence is the dangerous case. A check that only confirms what it expects to find will pass a trailer with no seal, a seal of a type it was not built to read, or a vehicle whose marking it could not read at all. Every check in this framework is required to report the negative result, and a missing or unreadable result is never treated as a pass.

Two independent signals, and a person decides. No single check releases or holds a load. Identity and integrity are checked separately, the system recommends, and a named person makes the release decision with the evidence in front of them. This is the rule that made automation acceptable to the investigators who had to trust it, and it is the rule that keeps a false reading from becoming a wrongful hold.

Record so that others can learn. A verification that leaves no record protects one load. A verification that leaves a structured record protects the next facility, the insurer, the investigator, and the agency trying to measure the problem. The evidence record is specified so that it can be shared, in anonymized form, across organizations.

5. Requirements

5.1 General

G-1 A party that releases or receives truckload freight shall have a written custody verification procedure that identifies, for each stage in Sections 5.2 to 5.4, who performs each check, what evidence is retained, and who may decide a verdict. [Basic]

Note. A one-page procedure is enough at the Basic level. The point is that the checks are assigned to someone and are not left to whoever happens to be at the gate.

G-2 The procedure shall name the records that establish the carrier of record for a load (for example the rate confirmation and the bill of lading) and shall require that the carrier of record be known to the releasing party before the presenting vehicle arrives. [Basic]

Note. A facility that learns who is picking up only when the driver says so has no carrier of record to verify against. This is the condition at the boundary between organizations that deceptive pickups exploit, and it is the first thing to fix.

G-3 The releasing party shall not accept the presenting vehicle's driver, paperwork, or telephone call as the sole source of the carrier of record. [Basic]

G-4 Verification checks shall be performed before the load leaves the releasing party's control, not after. [Standard]

Note. Checking the seal after the trailer has been dropped in the yard, which is common because it keeps the gate moving, verifies the past, not the handoff. Automated capture at the gate (Section 10) is how a facility meets this requirement without adding queue time.

G-5 Where any required check cannot be performed, or returns no result, the verifier shall record the fact and the reason, and the absence of a result shall not be treated as a pass. [Basic]

G-6 A procedure may use a vendor-specific seal, reader, camera, or software for any check, but it shall also define how the same check is completed when that device returns no result, is not fitted, or meets a seal or vehicle it cannot read, and the fallback shall produce the same evidence record as the primary method. [Basic]

Note. The common pattern at large networks is an RFID reader at the gate with a yard agent sent to read and record the seal number by hand when the reader fails or the seal carries no tag. That pattern conforms, provided the fallback is written into the procedure and its results land in the same record. What does not conform is a procedure that can verify only one make of seal and has nothing to say about every other trailer that arrives.

5.2 Pickup

Identity

P-1 Before releasing a load, the releasing party shall observe the carrier identity marking on the presenting vehicle and record the USDOT number displayed. [Basic]

Note. At Basic level this is a person reading the cab door and writing the number down or photographing it. At Standard and Full levels it is a camera capture and an automated read (P-5).

P-2 The releasing party shall resolve the recorded USDOT number against the registry and shall record the legal name, operating status, and out-of-service status returned. [Basic]

Note. FMCSA's public query service returns, among other elements, the carrier's legal name, DBA name, whether the carrier is allowed to operate, whether it is under an out-of-service order and the date, and its authority status. A lookup takes seconds and costs nothing.

P-3 The releasing party shall compare the legal or operating name returned by the registry with the carrier of record and shall record whether they match. [Basic]

Note. A match means the registry name and the carrier of record refer to the same company. Minor differences in punctuation, abbreviation, or a DBA name are not mismatches. A different company is a mismatch regardless of how similar the names look, since look-alike names are a known fraud technique.

P-4 The releasing party shall photograph the presenting vehicle so that the carrier identity marking and the license plate are legible, and shall retain the photographs in the evidence record. [Standard]

Note. This is the FBI's own recommendation: keep photographs of vehicles, license plates, cab numbers, and DOT and MC numbers. The photographs are what make an investigation possible after a fraudulent pickup.

P-5 The carrier identity marking shall be read automatically from the photograph, the read shall be resolved against the registry without manual transcription, and a read that cannot be resolved to a single USDOT number shall be reported as unreadable rather than guessed. [Full]

Note. This is the requirement that most facilities cannot meet today, and that is why it sits at the Full level rather than lower. It has been done at scale: a camera-based read of the USDOT number at gate check-in, resolved against the booked carrier, has run in production in a national network since 2025. Vendors of gate systems do not generally offer it, which is the reason the reference implementation in Section 10 is open. Automated reads should correct the usual confusions (O and 0, I and 1, S and 5) before resolving, and should emit one candidate or none. A system that emits its best guess when uncertain will eventually release a load to the wrong carrier on the strength of its own error.

P-6 Where the load was engaged through a broker or other intermediary, the releasing party should confirm the carrier of record with the intermediary through a channel other than the one on which the pickup was arranged. [Standard]

Note. A spoofed email or a cloned domain arranges the pickup; a telephone call to the broker's number on file, not the number in the email, confirms it.

P-7 The releasing party should record the driver's name and license state as shown on a government-issued identification presented in person, and should not accept a photograph of an identification document in place of the document. [Standard]

Integrity

P-8 When a seal is applied at pickup, the applying party shall record the seal identifier, the seal type, the time, and the identity of the person applying it, and shall provide the identifier to the carrier of record and the consignee. [Basic]

Note. This is the seal log that customs and industry guidance already require. The framework adds nothing new here except that the log is part of the evidence record.

P-9 Before the loaded trailer leaves the releasing party's control, a verifier shall observe that a seal is physically affixed to the door closure and shall record the observation. [Basic]

Note. The customs method is view, verify, tug, and twist: look at the seal, check the number against the documents, pull on it, and turn it. The framework requires at least the first two at Basic level.

P-10 The releasing party shall photograph the sealed door closure so that the seal and, where legible, its identifier are visible, and shall retain the photograph in the evidence record. [Standard]

P-11 The presence or absence of a seal on the door closure shall be determined from the photograph by a detector that reports seal present, seal absent, or uncertain, for seals of any type, and an uncertain result shall be treated as absent for the purpose of the verdict. [Full]

Note. A detector that reports only presence will pass a trailer with no seal. The "absent" class is the one that matters, and it is the one that a managed vision service cannot usually be trained to represent, which is why the reference implementation uses a purpose-trained model.

P-12 Where an electronic or RFID seal is used, the reader result shall be treated as a check of the seal identifier only, and shall not substitute for the observation or photograph required by P-9 and P-10. [Standard]

Note. An RFID reader confirms that a tag with the expected number is within range. It does not confirm that the seal is closed on the door, and it returns nothing at all for a mechanical seal or for no seal. The two checks answer different questions and the framework requires both.

Decision

P-13 The releasing party shall record a verdict for every pickup in accordance with Section 7 before the load leaves its control, and shall record the name of the person who decided it. [Basic]

P-14 A load with a verdict of hold shall not be released until the condition causing the hold has been resolved and the resolution recorded, or until a named person with authority to override has recorded the override and the reason. [Basic]

5.3 Intermediate stops

An intermediate stop is any point between pickup and final delivery at which the trailer doors may be opened, the load may be added to or partly unloaded, or custody passes between drivers or carriers: a multi-stop delivery route, a relay point, a cross-dock, a drop yard, or a transfer between tractors. Most of the loss from a seal cut early in a route happens after the cut, because freight keeps being loaded onto a trailer that is already open, and because the cut is not discovered until final delivery. The purpose of this section is to make the discovery happen at the next stop instead.

S-1 On arrival at an intermediate stop, before the doors are opened or any freight is handled, a verifier shall observe whether a seal is affixed and shall compare its identifier with the identifier recorded at the previous seal event. [Basic]

S-2 The verifier shall photograph the door closure on arrival, before the doors are opened, and the photograph shall be retained in the evidence record linked to the previous stop's record. [Standard]

S-3 The arrival photograph shall be assessed by the detector required by P-11, and the result shall be compared automatically with the result recorded at the previous seal event. [Full]

S-4 A missing seal, a seal whose identifier does not match the previous record, or a seal that shows evidence of tampering shall produce a verdict of hold, and no freight shall be loaded onto or removed from the trailer until the hold is resolved. [Basic]

S-5 When the doors are opened at an intermediate stop for a legitimate purpose and a new seal is applied, the stop shall record a seal event in accordance with P-8, and the new identifier shall be communicated to the carrier of record, the next stop, and the consignee before departure. [Basic]

S-6 Where custody passes between carriers or drivers at a stop, the identity requirements P-1 to P-4 shall apply to the vehicle and driver taking custody, as if the stop were a pickup. [Standard]

S-7 A carrier should perform the check in S-1 at any stop of its own, including driver changes and overnight parking, and should record the result, even where the stop is not a custody transfer. [Standard]

Note. A carrier that checks its own seal at each stop can show, after a theft, exactly where the loss occurred. That evidence is worth more to the carrier than to anyone else, since the carrier is usually the party held responsible.

Open question for comment: drop-and-hook. Many networks release loaded trailers into a yard where a different tractor, sometimes hours later and sometimes from a different carrier, hooks and leaves. This draft treats the hook as a custody transfer to which S-6 applies, but it does not say who verifies the second tractor when the yard is unattended, or how the seal check in S-1 is performed when no one is present at the hook. Readers who run drop yards are asked to say how this is handled today and what a workable requirement would look like. The author's current view is that an unattended hook should be photographed by a fixed camera and resolved under P-5, which would place unattended drop-and-hook at the Full level only.

5.4 Delivery

D-1 On arrival at the consignee, before the doors are opened, a verifier shall observe whether a seal is affixed and shall compare its identifier with the identifier most recently communicated under P-8 or S-5. [Basic]

D-2 The consignee shall photograph the door closure on arrival, before the doors are opened, and the photograph shall be retained in the evidence record. [Standard]

D-3 The consignee shall observe and record the carrier identity marking on the delivering vehicle and should resolve it against the registry in accordance with P-2 and P-3. [Standard]

Note. Verification at delivery catches the diverted load that was picked up legitimately and then re-brokered: the vehicle that arrives is not the carrier of record. It also closes the loop for the shipper, who otherwise learns of the diversion only when the carrier of record asks why it was never paid.

D-4 A missing seal, a mismatched identifier, or a mismatched carrier at delivery shall be recorded as an incident in the evidence record, shall be reported to the carrier of record and the shipper or broker before the freight is accepted, and should be reported to the relevant theft-recording service and, where a loss is found, to law enforcement. [Basic]

D-5 The consignee shall record the removal of the seal as a seal event, including who removed it and when. [Basic]

6. The evidence record

The evidence record is the set of data and images retained for each custody transfer. It has two purposes. The first is operational: it is what a releasing party looks at before deciding, and what an investigator looks at after a loss. The second is collective: records kept in a common structure can be shared, anonymized, and aggregated, which is how the industry and its regulators will eventually measure a problem that today is counted differently by every tracker. The field list below is the minimum. Parties may add fields. The structure is published in machine-readable form with the reference implementation.

E-1 A releasing party shall retain, for each custody transfer, the fields marked Basic in Table 6.1, for not less than one year or the period required by its insurer or contracts, whichever is longer. [Basic]

E-2 A releasing party shall retain the fields marked Standard in Table 6.1, including image references, and shall be able to produce the images on request by the carrier of record, the shipper, the insurer, or law enforcement. [Standard]

E-3 Evidence records shall be stored in the published structure, or in a structure from which it can be produced without manual re-entry, so that records can be exported for sharing. [Full]

E-4 Personal data in the evidence record (driver name, license details, faces visible in images) shall be retained only as long as needed for the purposes above and shall be excluded or redacted from any record shared outside the organization unless the sharing is to law enforcement or is required by law. [Basic]

Table 6.1. Evidence record fields.

Field	Content	Level
Transfer identifier	Unique identifier for this custody transfer	Basic
Stage	Pickup, stop, or delivery	Basic
Timestamp	Date and time of the verification, with time zone	Basic
Site	Identifier of the releasing or receiving site	Basic
Load reference	Bill of lading, order, or load number	Basic
Carrier of record	USDOT number and legal name from the tender or rate confirmation	Basic
Source of carrier of record	Which document or system established it, and whether it was confirmed with the intermediary (P-6)	Standard
USDOT as read	The number observed or read from the presenting vehicle	Basic
Read method	Manual, photograph plus manual, or automated	Standard
Registry result	Legal name, DBA name, allowed to operate, out-of-service flag and date, authority status, as returned	Basic
Identity match	Match, mismatch, or unresolved, with the reason for any mismatch	Basic
Red flags	Any rule in Section 7.2 that fired, by identifier	Standard
License plate	As observed or read	Standard
Driver	Name and license state as presented (subject to E-4)	Standard
Seal identifier expected	From the previous seal event or the shipping documents	Basic
Seal observed	Present, absent, or uncertain; identifier if legible; seal type	Basic
Seal detection method	Manual observation, photograph plus manual, or automated detector	Standard
Seal match	Match, mismatch, absent, or unreadable	Basic
Previous seal event	Reference to the prior record on multi-stop routes	Standard
Images	References to the cab-side and door-closure photographs, with capture time	Standard
Verdict	Release, verify, or hold, and the reasons	Basic
Decided by	Name or identifier of the person who decided	Basic
Override	Whether a hold was overridden, by whom, and why	Basic
Incident	Whether an incident was recorded, and to whom it was reported	Basic

7. Verdicts and risk rules

7.1 Verdict grading

A verdict is the graded outcome of the identity and integrity checks taken together. Three grades are enough. More than three invites argument at the gate about which one applies; fewer than three forces a choice between stopping every doubtful load and stopping none.

V-1 A verdict of release shall be recorded only when the identity check and the integrity check have both been performed and both returned a match, and no red flag in Section 7.2 has fired. [Basic]

V-2 A verdict of verify shall be recorded when any check returned an unresolved, uncertain, or unreadable result, or when a red flag classified as advisory fired, and the load shall not be released until a person has completed the check by another means and recorded the result. [Basic]

Note. Verify is the grade for "the system could not tell." A bad photograph, a cab whose number is painted over, a registry outage, and a stale filing all land here. The load is not accused of anything; it is asked to wait for a person.

V-3 A verdict of hold shall be recorded when the identity check returned a mismatch, when the integrity check returned absent or mismatch, or when a red flag classified as blocking fired, and the load shall not be released except under P-14. [Basic]

V-4 The verdict shall be recommended by the procedure or system and decided by a named person, who shall have the evidence record, including any images, available at the time of decision. [Basic]

V-5 A party that automates any check shall measure, at least quarterly, the rate at which automated results were overridden by the deciding person and the rate at which loads released on an automated match were later the subject of an incident, and shall review the thresholds of the automation against those rates. [Standard]

Note. This is how a facility knows whether its automation is helping. In my own deployment the measures that mattered were the share of flagged loads safely deprioritized and the number of later-confirmed thefts among them, and both were reviewed before the automation was allowed to affect decisions.

7.2 Risk rules

A risk rule is a condition, evaluated against the registry result and the booking, that changes the verdict even when names match. The rules below are drawn from the FBI's public guidance on cyber-enabled cargo theft, from FMCSA's registration enforcement, and from patterns observed in production fraud cases. Each is classified as blocking (produces hold) or advisory (produces verify). The classification is a default; a party may reclassify a rule in its procedure and shall record the reason.

Rule	Condition	Default	Basis and notes
R-1	The USDOT number on the vehicle resolves to a different legal entity than the carrier of record	Blocking	The signature of a deceptive pickup and of unlawful sub-brokering. Look-alike names are a mismatch.
R-2	The registry shows the carrier is not allowed to operate, or is under an out-of-service order	Blocking	Registry element "allowToOperate" and the out-of-service flag and date.
R-3	The carrier's operating authority is inactive, revoked, or pending for the type of operation	Blocking	Authority status from the registry's authority endpoint.
R-4	No seal is present where one was expected, or the identifier does not match the last recorded seal event	Blocking	Section 5 integrity checks.

Rule	Condition	Default	Basis and notes
R-5	The registry's most recent carrier filing (MCS-150) is more than two years old	Advisory	Stale registrations are common among dormant authorities purchased for fraud. FMCSA's March 2026 bulletin prohibits buying or selling USDOT numbers outside a legitimate corporate transaction.
R-6	The registry records zero power units or zero drivers for a carrier presenting with a tractor	Advisory	A paper carrier that owns no equipment cannot lawfully be the one hauling.
R-7	The carrier's authority is less than 90 days old and the load value exceeds the party's stated threshold	Advisory	New authorities are not fraudulent by nature, but new authorities with high-value loads are where losses concentrate. The threshold is set by each party.
R-8	The registry's physical address or telephone matches an address or telephone previously associated with a carrier placed out of service	Advisory	The chameleon-carrier pattern that FMCSA's registration reforms target. Requires a party to keep its own history or to use a service that does.
R-9	The pickup was arranged, or the carrier of record was changed, through a channel that could not be confirmed under P-6	Advisory	The FBI's guidance centers on phished credentials and spoofed communications; an unconfirmed change of carrier is the most common precursor.
R-10	The presenting vehicle's license plate has been recorded at this or an affiliated site under a different USDOT number within a period set by the party	Advisory	The same tractor appearing under successive identities is the physical trace of both sub-brokering and reincarnated carriers.

R-11 A party shall document which rules in this section it evaluates, with what data source, and shall record in the evidence record each rule that fired. [Standard]

R-12 A party should not implement any rule in this section as a filter that removes the load from review. A rule that fires shall change the verdict, not hide the load. [Standard]

Note. In production, a hard filter on "known carrier" would have discarded exactly the double-brokered loads the system existed to find. Rules adjust confidence; they do not decide.

8. Conformance levels

A party may claim conformance to this framework at one of three levels. The levels are cumulative. They are designed so that the Basic level requires no equipment and no software beyond a telephone and a web browser, so that a facility of any size can begin, and so that the Full level can be met with the cameras most facilities already own and open-source software.

Level	What it means in practice	Requirements
Basic	A written procedure; the carrier of record known before arrival; the USDOT number read from the vehicle by a person and resolved against the registry; the seal observed and its identifier compared; a verdict recorded with the decider's name; the Basic fields of the evidence record retained.	G-1 to G-3, G-5, G-6, P-1 to P-3, P-8, P-9, P-13, P-14, S-1, S-4, S-5, D-1, D-4, D-5, E-1, E-4, V-1 to V-4
Standard	Basic, plus photographs of the cab and the door closure retained for every transfer; verification before release rather than after; confirmation of brokered carriers through a second channel; identity checks at custody-changing stops and at delivery; red flags evaluated and recorded; automation measured where used.	All Basic, plus G-4, P-4, P-6, P-7, P-10, P-12, S-2, S-6, S-7, D-2, D-3, E-2, V-5, R-11, R-12

Level	What it means in practice	Requirements
Full	Standard, plus automated reading of the carrier marking resolved against the registry; automated seal detection reporting present, absent, or uncertain for any seal type; automated comparison with the previous stop; evidence records in the published structure.	All Standard, plus P-5, P-11, S-3, E-3

C-1 A party claiming conformance shall state the level claimed, the sites or operations to which the claim applies, and the version of this document, and shall be able to produce its written procedure and a sample of evidence records on request. [Basic]

C-2 A party shall not claim conformance for a site at which any requirement of the claimed level is not met, but may claim a lower level for that site. [Basic]

C-3 A vendor claiming that a product supports conformance shall state which requirements the product performs and which remain the responsibility of the party's procedure. [Standard]

Note. No product performs G-1 or V-4. A product that claims to "make a facility conformant" is misdescribing itself.

Self-assessment checklist. Appendix A reproduces the requirements as a checklist with a column for each level. A party should complete it before claiming a level and keep the completed checklist with its procedure. This version of the framework has no certification scheme. If associations or insurers wish to audit against it, the checklist is the audit instrument, and the author will assist in adapting it.

9. Relationship to existing guidance and standards

This framework is written to sit between documents that already exist, not to replace them. The table below maps its requirements to the guidance they implement or extend, so that a party already following one of those documents can see what is new.

Document	What it provides	How this framework relates
FBI IC3 public service announcement I-043026, "Cyber-Enabled Strategic Cargo Theft Surging" (April 30, 2026)	Recommends independently verifying shipment requests and pickups by secondary methods before release; keeping photographs of drivers, licenses, vehicles, plates, cab numbers, and DOT and MC numbers; multi-channel verification.	G-3, P-4, P-6, P-10, R-9 make the recommendations into requirements with a record. The framework is, in effect, the implementation the announcement asks for.
U.S. DOT Request for Information, Docket DOT-OST-2025-1326 (September 2025)	Asks which technologies (including electronic seals and AI-driven monitoring) have proven effective; what barriers prevent timely detection, reporting, and response; what data collection improvements and integrations with FMCSA data are needed.	Sections 5 and 10 answer the technology question; G-4 and Section 5.3 the detection-timing question; Section 6 the data question. The evidence record is designed to be aggregated.
FMCSA registration system (Motus, 2026) and bulletin on USDOT number transfers (March 13, 2026)	Verifies the identity of company officials at registration; prohibits buying, selling, or leasing USDOT numbers outside a legitimate corporate transaction. Does not verify drivers or physical pickups.	P-1 to P-5 and R-1 to R-3 are the physical-pickup verification that registration reform stops short of; R-5, R-8, R-10 target purchased and reincarnated identities.
CBP CTPAT seal procedures and ISO 17712	Requires high-security seals for containers in the program; the view, verify, tug, twist inspection; a seal log; comparison of seal number with shipping documents; replacement and notification procedures	P-8, P-9, S-1, S-5, D-1, D-5 adopt the same seal discipline for domestic truckload freight regardless of seal class, and add photographs and automated detection (P-10, P-11) that CTPAT does not require.

Document	What it provides	How this framework relates
	when a seal is compromised.	
TAPA Facility Security Requirements and Trucking Security Requirements (2023)	Certifiable physical security standards for facilities and vehicles at three levels; include seal and documentation controls at handover.	This framework specifies the verification transaction, not the facility or vehicle. A TAPA-certified facility meets several Basic requirements already; the identity resolution against the registry (P-2, P-3) and the evidence record (Section 6) are additions. [Clause-level cross-references to be added after review against the full FSR and TSR text.]
TIA Framework to Combat Fraud (July 2024) and State of Fraud reports	Describes fraud categories and recommends carrier vetting at booking and diligent communication; states that it is not intended to be a checklist or standard.	This framework begins where booking-time vetting ends. P-6 and R-9 connect the two; the evidence record gives brokers the pickup-side facts they currently lack when a load disappears.
49 CFR 390.21 (marking of commercial motor vehicles)	Requires the legal or trade name and USDOT number to be displayed on both sides of the power unit, legible from 50 feet in daylight.	The marking is the physical anchor for P-1 and P-5. A vehicle without a legible marking cannot receive a verdict of release (G-5, V-2).

10. Reference implementation and evaluation

A framework that could only be followed on paper would not survive contact with a busy gate. This section describes open-source software, released under the Apache License 2.0, that implements the automated requirements, and the evaluation set that measures whether it does so well enough. The software is one implementation among possible others; conformance is to the requirements, not to the software.

carrier-verify (released; Python Package Index, version 0.1.0, September 10, 2026) implements P-2, P-3, and the registry-based rules R-1 to R-3 and R-5 to R-7. It looks up a carrier by USDOT number, name, or docket through FMCSA's public query service, evaluates the red-flag rules, compares the resolved identity with the carrier of record on the rate confirmation, and returns a verdict of green, yellow, or red (corresponding to release, verify, and hold) with the reasons. It accepts raw text from an optical character reader and corrects the usual confusable characters before resolving (P-5), and it exits with codes that gate automation can act on.

seal-check [release date and version to be inserted] implements P-11 and S-3. It is a detector trained to report seal present, seal absent, or uncertain on a photograph of a trailer door closure, for bolt, cable, plastic indicative, padlock, and electronic seals. Version 0.1 was trained on photographs of purchased seals on a door mock-up under daylight, lamp, rain, and glare, and on synthetic renders; it was not trained on any proprietary imagery. Its evaluation results on the SealBench-Lite set are reported in the release notes and should be read as a baseline for the community to improve, not as the ceiling of the method. [Insert the evaluation table: images, seal types, detection rate, "no seal" accuracy by condition.]

gate-verify [release date and version to be inserted] implements the pipeline: it takes a cab-side image and a door-closure image, runs the identity read and carrier-verify, runs seal-check, compares with the previous stop's record where one is supplied, produces a verdict with reasons, and writes an evidence record in the structure of Section 6. It runs on a laptop and calls only public data. It performs no requirement in Sections 5.1 or 7.1 that calls for a person; it recommends, and the deciding person is prompted for a name.

SealBench-Lite [release date and version to be inserted] is the evaluation set released with seal-check: the seal taxonomy, the collection protocol with consent and redaction rules, the synthetic generator, and the labeled

images. It is the seed of a larger benchmark with consented field collection at multiple facilities, planned for 2027 and 2028, and it exists so that any detector, commercial or open, can be measured against the same images for the “absent” case that matters most.

Provenance. The methods in this section descend from systems the author designed, evaluated, and deployed in a national trucking network between 2025 and 2026, including a camera-based carrier identity read at gate check-in and a seal-detection model enabled across that network in May 2026. The open implementations share no code, data, or imagery with those systems. They were written from the public registry, from purchased seals, and from the requirements in this document.

11. Versions, changes, and how to comment

Versions are numbered major.minor. A minor version adds requirements or notes, clarifies wording, or changes conformance level tags; it does not withdraw a requirement or change its meaning. A major version may do either. Every change is listed in a change log with the identifier affected and the reason. Proposed changes are posted for comment in the repository for at least 30 days before a new version is published, and a requirement is never changed and published in the same step.

Comments are welcome from anyone, and especially from people who run gates. The most useful comment is one that says “at my facility this requirement would fail because” followed by the reason. Comments that are adopted are credited in the change log unless the commenter asks otherwise. The document is maintained by its author; if an association, council, or standards body wishes to take over maintenance, the license permits it and the author will support the transfer.

[Contact details, repository URL, and comment deadline for version 0.1 to be inserted.]

Appendix A. Self-assessment checklist

Mark each requirement met, not met, or not applicable, and attach the reason for any requirement not met. A level is claimed only when every requirement at that level and below is met for the sites named.

ID	Requirement (short form)	Level	Met?
G-1	Written procedure assigning checks, evidence, and deciders	Basic	
G-2	Carrier of record known before arrival from named records	Basic	
G-3	Driver, paperwork, or call not the sole source of carrier of record	Basic	
G-4	Checks performed before release, not after	Standard	
G-5	Missing results recorded; never treated as pass	Basic	
G-6	No dependence on a single vendor’s seal, camera, reader, or software	Basic	
P-1	USDOT number observed on vehicle and recorded	Basic	
P-2	Number resolved against registry; status recorded	Basic	
P-3	Registry name compared with carrier of record	Basic	
P-4	Cab and plate photographed and retained	Standard	
P-5	Automated read, resolved without transcription; unreadable reported	Full	
P-6	Brokered carrier confirmed through second channel	Standard	

ID	Requirement (short form)	Level	Met?
P-7	Driver identification recorded from document in person	Standard	
P-8	Seal application logged and communicated	Basic	
P-9	Seal observed before release	Basic	
P-10	Door closure photographed and retained	Standard	
P-11	Automated present / absent / uncertain detection, any seal type	Full	
P-12	RFID result not a substitute for observation	Standard	
P-13	Verdict recorded with decider	Basic	
P-14	Holds resolved or overridden with record	Basic	
S-1	Seal observed and compared on arrival at stops	Basic	
S-2	Door closure photographed on arrival at stops	Standard	
S-3	Automated comparison with previous stop	Full	
S-4	Hold on missing, mismatched, or tampered seal; no handling until resolved	Basic	
S-5	Re-seal logged and communicated before departure	Basic	
S-6	Identity checks at custody-changing stops	Standard	
S-7	Carrier self-checks at own stops	Standard	
D-1	Seal observed and compared on arrival at consignee	Basic	
D-2	Door closure photographed at delivery	Standard	
D-3	Delivering vehicle’s marking recorded and resolved	Standard	
D-4	Incidents recorded and reported before acceptance	Basic	
D-5	Seal removal logged	Basic	
E-1	Basic fields retained one year or longer	Basic	
E-2	Standard fields and images retained and producible	Standard	
E-3	Records in or exportable to the published structure	Full	
E-4	Personal data limited and redacted when shared	Basic	
V-1	Release only on two matches and no blocking flag	Basic	
V-2	Verify on unresolved results or advisory flags	Basic	
V-3	Hold on mismatch, absence, or blocking flag	Basic	
V-4	Named person decides with evidence available	Basic	
V-5	Automation override and incident rates measured quarterly	Standard	
R-11	Rules evaluated are documented with data sources	Standard	
R-12	No rule implemented as a filter	Standard	
C-1	Claim states level, sites, version; procedure and samples producible	Basic	
C-2	No claim for a site that does not meet the level	Basic	
C-3	Vendor claims state which requirements the product performs	Standard	

Appendix B. Sources

[1] Federal Bureau of Investigation, Internet Crime Complaint Center, Public Service Announcement I-043026, “Cyber-Enabled Strategic Cargo Theft Surging,” April 30, 2026. https://www.ic3.gov/PSA/2026/PSA260430

[2] U.S. Department of Transportation, “Protecting America’s Supply Chain From Cargo Theft: Request for Information,” Docket DOT-OST-2025-1326, 90 Fed. Reg. 45309, September 19, 2025.

[3] Federal Motor Carrier Safety Administration, QCMobile API documentation and response elements (allowToOperate, outOfService, outOfServiceDate, dotNumber, mcNumber, legalName, dbaName, physical address and telephone elements; carrier, docket, authority, and out-of-service endpoints). https://mobile.fmcsa.dot.gov/QCDevsite/docs/qcApi and https://mobile.fmcsa.dot.gov/QCDevsite/docs/apiElements

[4] Federal Motor Carrier Safety Administration, bulletin of March 13, 2026 on the sale, purchase, or lease of USDOT numbers and operating authority, as reported by FreightWaves. https://www.freightwaves.com/news/fmcsa-just-issued-a-bulletin-warning-carriers-not-to-buy-or-sell-dot-numbers-here-is-why-that-warning-exists-and-what-it-means

[5] U.S. Customs and Border Protection, “Recommended Seal Procedures Including Best Practices” (CTPAT): view, verify, tug, twist; ISO 17712 high-security seals; seal log; comparison of seal number with shipping documents; procedures for compromised seals. https://www.cbp.gov/sites/default/files/documents/Recommended%20Seal%20Procedures%20Including%20Best%20Practices.pdf

[6] International Organization for Standardization, ISO 17712:2013, Freight containers, mechanical seals (classes I, S, and H).

[7] Transported Asset Protection Association, Facility Security Requirements (FSR) 2023 and Trucking Security Requirements (TSR) 2023, certification framework and security levels 1 to 3. https://tapaemea.org/wp-content/uploads/2023/08/TSR-2023-Standard.pdf

[8] Transportation Intermediaries Association, “2024 TIA Framework to Combat Fraud” (July 31, 2024) and “State of Fraud in the Industry” (April 2025). https://news.tianet.org/2024-tia-framework-to-combat-fraud/

[9] Verisk CargoNet, 2025 annual theft trends (2,646 incidents; $725 million; average $273,990) and 2026 quarterly reports. https://www.cargonet.com/news-and-events/cargonet-in-the-media/2025-theft-trends/

[10] Overhaul, Q2 2026 cargo theft report (pilferage 46% of incidents; 68% of Southern California corridor incidents within the first 50 miles of origin), as reported by Truck News, August 13, 2026.

[11] Testimony of Donna Lemm on behalf of the American Trucking Associations before the U.S. Senate Committee on the Judiciary, July 15, 2025 (strategic theft up more than 1,500% since Q1 2021; load boards and unknown counterparties).

[12] 49 CFR 390.21, Marking of self-propelled CMVs and intermodal equipment.

[13] IETF RFC 2119, “Key words for use in RFCs to Indicate Requirement Levels,” and RFC 8174.

[14] RFID Journal, “RFID-Enabled Electronic Seals” (seal costs, reader infrastructure, limits of passive e-seals).

Appendix C. Change log

Version	Date	Change
0.1 draft 1	October 5, 2026	First draft.
0.1 draft 2	October 6, 2026	Deposited on Zenodo, DOI 10.5281/zenodo.23179115. G-6 reworded after gate review: vendor-specific devices permitted with a documented fallback producing the same record. P-5 note added on current feasibility and level. Open question on drop-and-hook added to Section 5.3.